Advanced Computer Graphics (ACG)

Theory Session 8 – Volume Rendering – Emission and Scattering

2026-2027

Pompeu Fabra University

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Calendar

Week no.	Date	Theory	Practice	Seminar	Deliveries
Week 1	28 Sep 02 Oct	**T1**: Introduction	**P1**: Introduction to A1.1	-	
Week 2	05 Oct 09 Oct	**T2**: Basic Ray Tracing		**S1**: Work on A1	
Week 3	12 Oct 16 Oct	**T3**: Ray Casting, Recursive Ray Tracing	**P2**: Introduction to A1.2		
Week 4	19 Oct 23 Oct	**T4**: Rendering Equation, Monte Carlo Integration and Direct Illumination **T5**: Indirect Illumination and Path Tracing		**S2**: Work on A1	**D.A1**
Week 5	26 Oct 30 Oct	**T6**: Volume Rendering: Absorption-Only Model	**P3**: Introduction to A2.1		
Week 6	02 Nov 06 Nov	**T7**: Volume Rendering: Emission, Absorption and Scattering Model **T8:** Neural Radiance Fields and Topics' Presentation		**S3**: Work on A2	
Week 7	09 Nov 13 Nov	**T9**: **Midterm**	**P4**: Introduction to A2.2		
Week 8	16 Nov 20 Nov		**P5**: Work on A2	**S4**: Work on A2	**D.A2**
Week 9	23 Nov 27 Nov	**T10**: **1st Presentation**	**P6**: Work on FP	-	
Week 10	30 Nov 04 Dec	**T11**: **2nd Presentation** **(First Session)** **T12**: **2nd Presentation** **(Second Session)**		**S5**: Work on FP	**D.FP**

Class Outline

- Recall Last Class
  - Radiative Transfer Equation
  - From the RTE to the Volume Rendering Equation (VRE)
    - The case of Absorption-Only
  - Approximating the Absorption-Only VRE

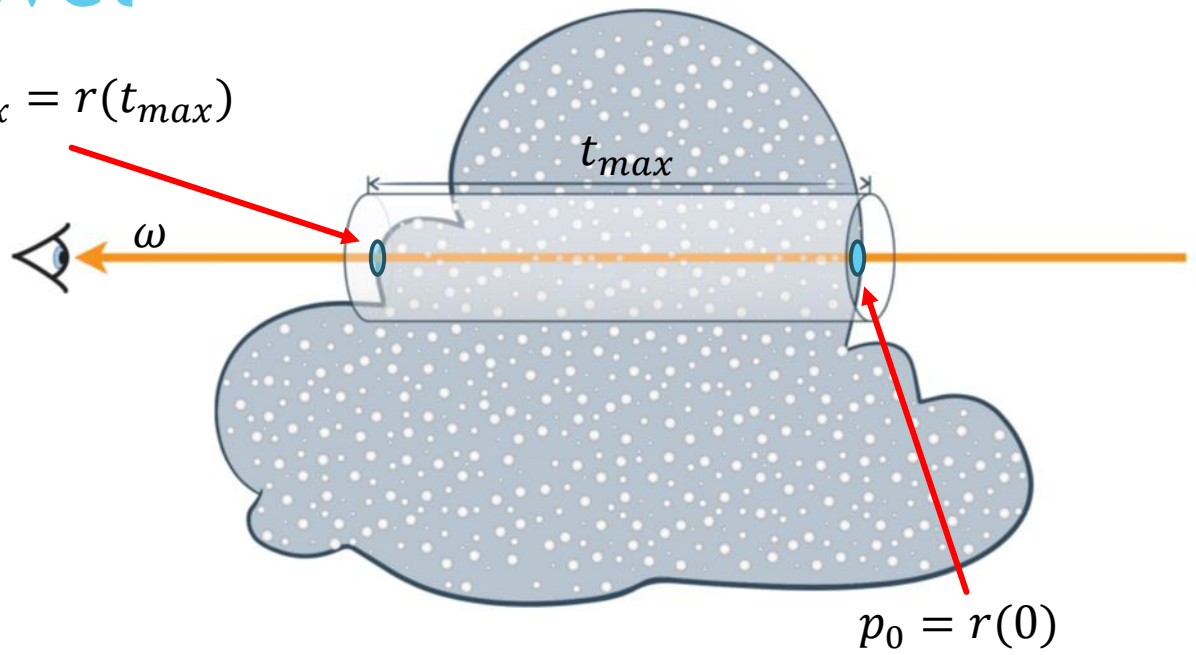
- Today (1): Absorption-Emission-Scattering
  - *Emission-Absorption* Model
  - Full Volume Rendering Model

- Today (2): Volume Representation and Sampling
  - 3D Textures
  - Linear, Bi-linear and Tri-linear interpolation

Recall

Last Class in 5 Slides

The Macro Level



$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) = -\mu_t(t)L(t) + \mu_s(t)L_s(t) + \mu_a(t)L_e(t)$$

- Gives us the change in radiance at each point in the ray
- We want radiance $L(t_{max})$ after interacting with all the volume (integral along the path)
- *Volume Rendering Equation*

Micro-View

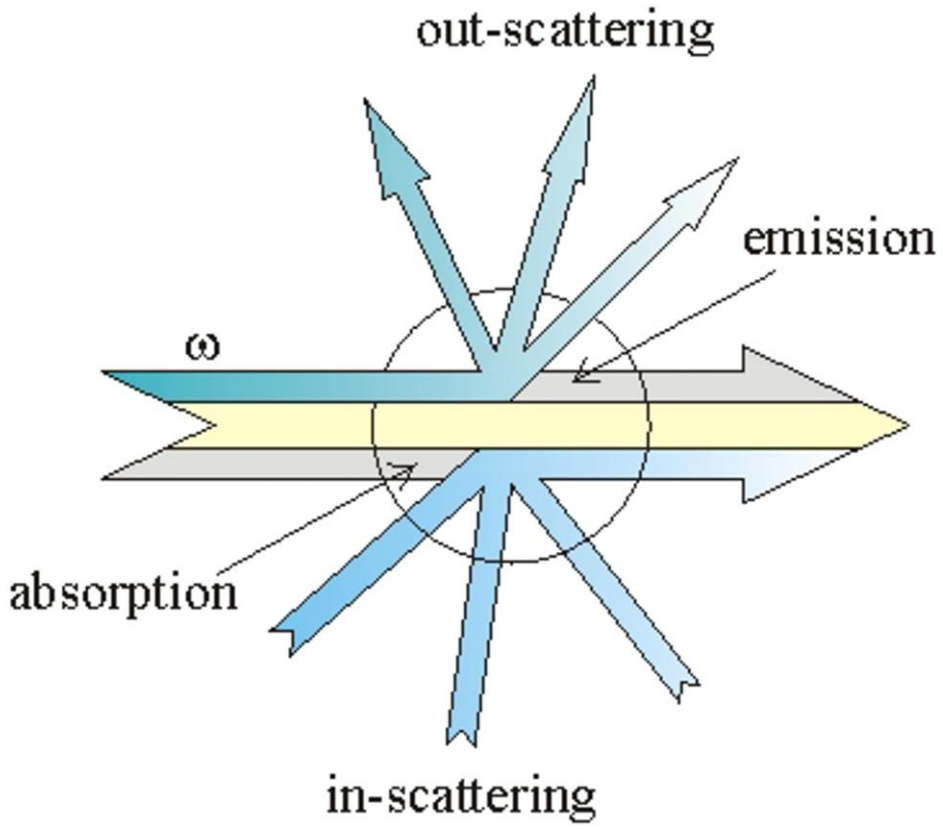


Image courtesy of Hendrik Lensch (MPI, WS07/08)

Understanding the Absorption Model

- The integral in the exponent is called ***optical thickness*** (τ)
  - $\tau(t_a, t_b) = \int_{t_a}^{t_b} \mu_a(t')\,\mathrm{d}t'$
  - Sums the attenuation μ_a coefficient from t_a to t_b
  - $\tau(t_a, t_b) \geq 0$
- The exponent term is called ***Transmittance*** (T)
  - $T(t_a, t_b) = e^{-\tau(t_a, t_b)}$ (Beer-Lambert Law)
  - $t_a \leq t_b$
  - Measures the fraction of radiance transmitted between t_a and t_b
  - $0 \leq T(t_a, t_b) \leq 1$

Different ways of writing the Absorption-Only VRE

$$L(t) = B\, e^{-\int_0^t \mu_a(t')\,\mathrm{d}t'}$$

$$L(t) = B e^{-\tau(0,t)}$$

$$L(t) = B\, T(0,t)$$

Ray Marching Loop for Absorption Model
Putting it all Together

Ray Set-up

Launch the viewing ray r

$t_{max} \leftarrow \text{intersectVolumeBoundaries}(r)$

Traversal Loop

$\tau = 0$

$\Delta_t = \text{computeStep}(\dots)$

$t = \Delta_t/2$

While $t < t_{max}$

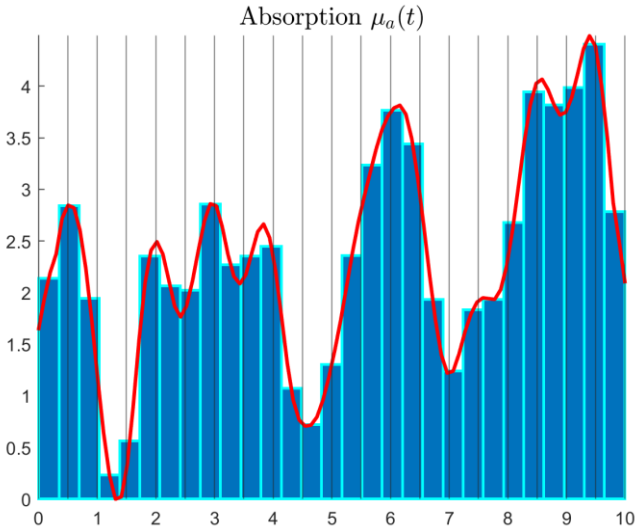
$\mu_a^t \leftarrow \text{getAbsorption}(t)$

$\tau \mathrel{+}= \mu_a^t \times \Delta_t$

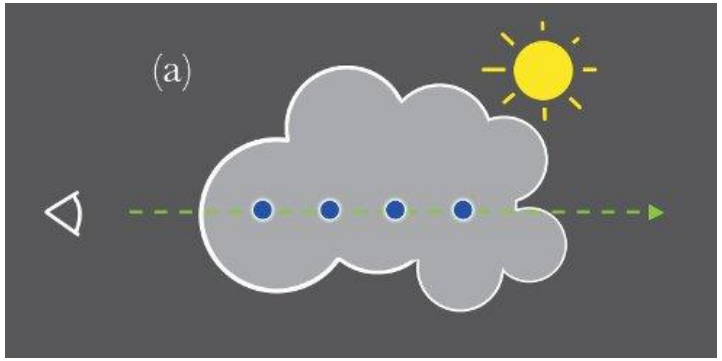
$t \mathrel{+}= \Delta_t$

Final Treatment

$\text{pixelColor} = B \exp(-\tau)$



$$\tau(t_a, t_b) = \int_{t_a}^{t_b} \mu_a(t')\, \mathrm{d}t'$$





Absorption Model Example Result Heterogeneous Volume

Emission-Absorption Model

Emission-Absorption Model Set Up

- Only emission and absorption are considered (no scattering)
  - $\mu_s(t) = 0$ ← (no scattering)
  - $\mu_t(t) = \mu_a(t) + \mu_s(t) = \mu_a(t)$
  - $L_e(t) \geq 0$

- The Radiative Transfer Equation is simplified to:

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) + \mu_a(t)L(t) = \mu_a(t)L_e(t)$$

- 1st Order ODE (non-homogeneous, rhs different from zero)

$$y'(x) + p(x)\,y(x) = q(x)$$

Solving 1st Order ODE (non-homogeneous)

- We'll be using the *Integrating Factor* technique, which consists of multiplying both sides of the ODE by the factor I, where:

$$I = e^{\int p\,dx} \rightarrow I' = p\,e^{\int p\,dx} = p\,I \quad (= pI)$$

$$\frac{dL}{dt}(t) + \mu_a(t)\,L(t) = \mu_a(t)\,L_e(t)$$

$$I = e^{\int_0^t \mu_a(t)\,dt}$$

$$I' = \mu(t)\,I(t) = \mu(t)\,e^{\int_0^t \mu_a(t)\,dt}$$

$$L'(t)\,I + \mu_a(t)\,I(t)\,L(t) = \mu_a(t)\,L_e(t)\,I$$

$$L'(t)\,I + I'\,L(t) = \mu_a(t)\,L_e(t)\,I$$

$$(I \cdot L(t))' = \mu_a(t)\,L_e(t)\,I \quad \text{INTEGRATE}$$

$$I(t)\,L(t) - I(0)\,L(0) = \int_0^t \mu_a(t)\,L_e(t)\,I(t)$$

$$L(0) = B$$

$$I(0) = 1$$

$$I(t)\,L(t) = B + \int_0^t \mu_a(t')\,L_e(t')\,I(t')\,dt'$$

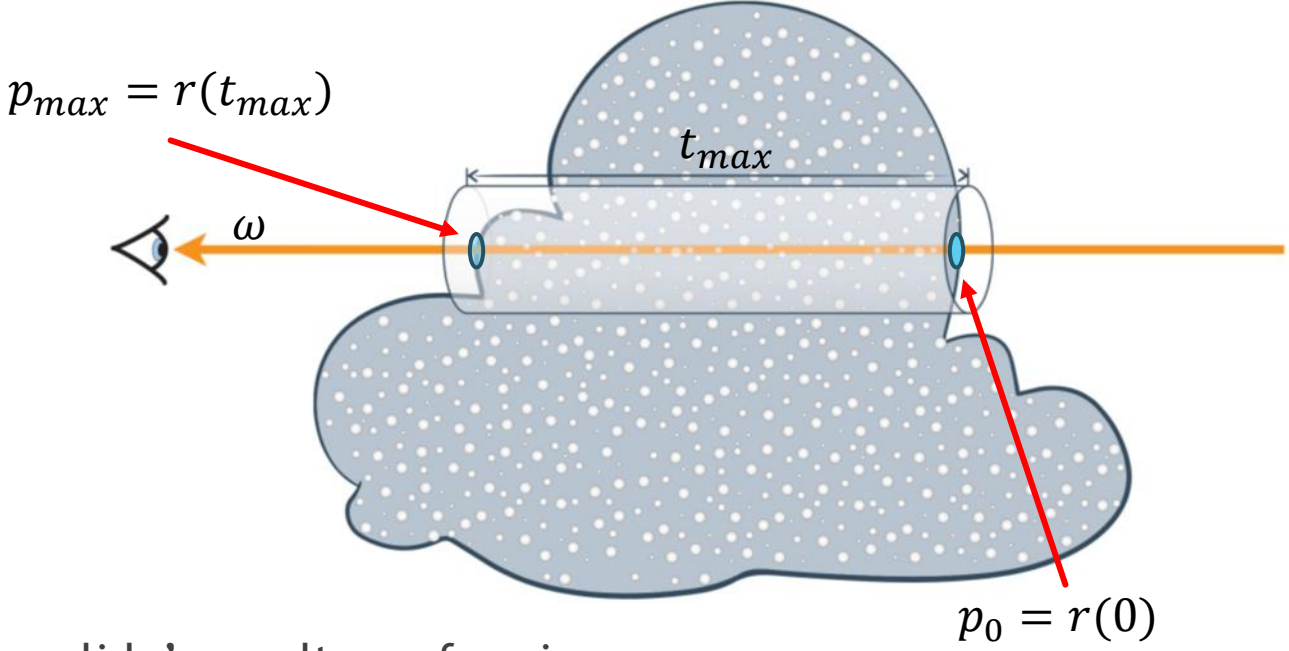
$$L(t) = \frac{B}{I(t)} + \int_0^t \mu_a(t')\,L_e(t')\,\frac{I(t')}{I(t)}\,dt'$$

$$\rightarrow \frac{1}{I(t)} = e^{-\int_0^t \mu_a(t)} = T(0,t) \qquad \left(\frac{e^0}{I(t)} = e^{0 - \int_0^t \mu_a(t)}\right)$$

$$\rightarrow \frac{I(t')}{I(t)} = e^{\int_0^{t'} \mu_a(t) - \int_0^t \mu_a(t)} = e^{\int_{t'}^t \mu_a(t)\,dt'} = T(t',t)$$

$$L(t) = \int_0^t \mu_a(t')\,L_e(t')\,T(t',t)\,dt' + B\,T(0,t)$$

Back to the Emission-Absorption Model



- Replacing the respective functions in the previous slide's result, performing detailed definite integrals computations and using the boundary condition $L(0) = B$ to find the constant C, we get:

$$T(t_a, t_b) = e^{-\int_{t_a}^{t_b} \mu_a(t')\,\mathrm{d}t'}$$

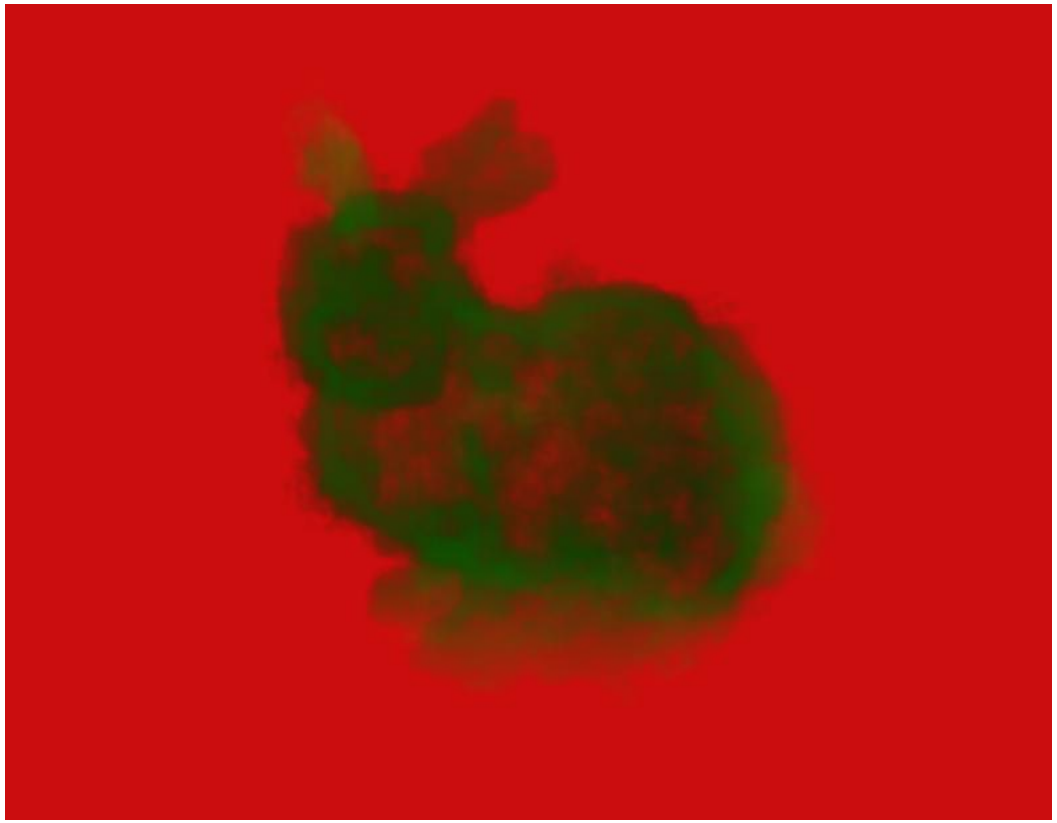
$$L(t) = \int_0^t \mu_a(t')\, L_e(t')\, T(t', t)\, dt' + B\, T(0, t)$$

Light emitted at the current point t' attenuated by μ_a

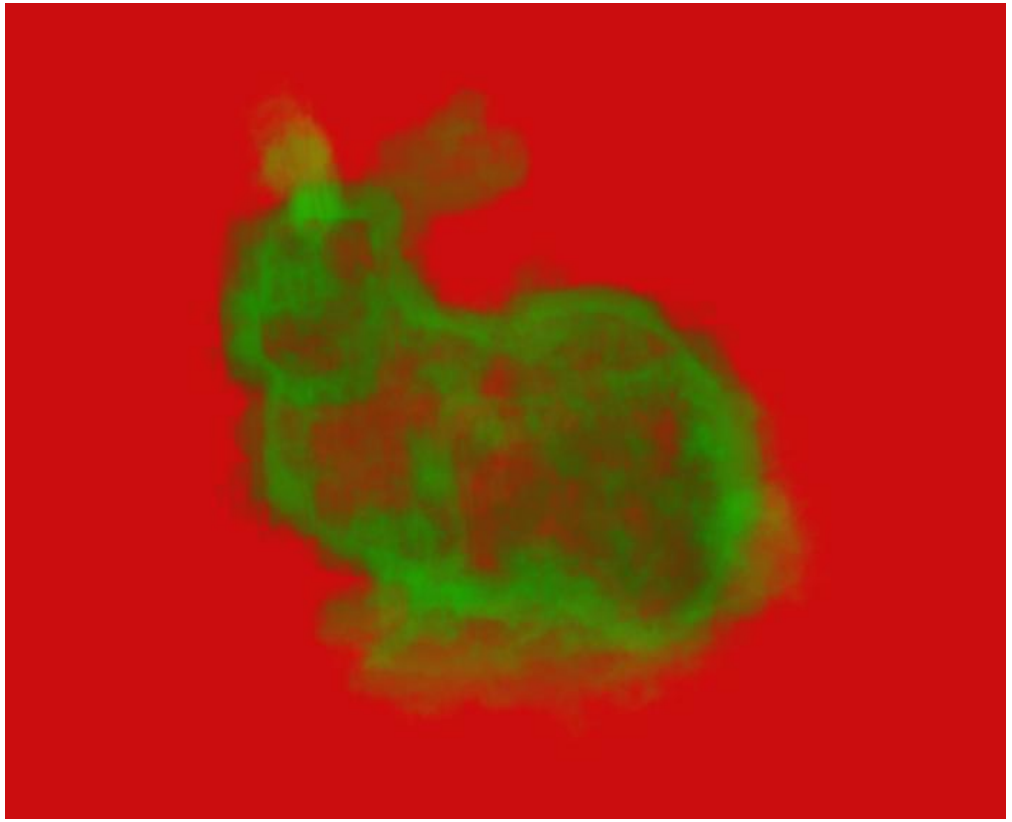
Attenuation of the light emitted at t' till it reaches t

Attenuated background color

Absorption = 0.02, Ambient light color intensity x20, Heterogeneous volume



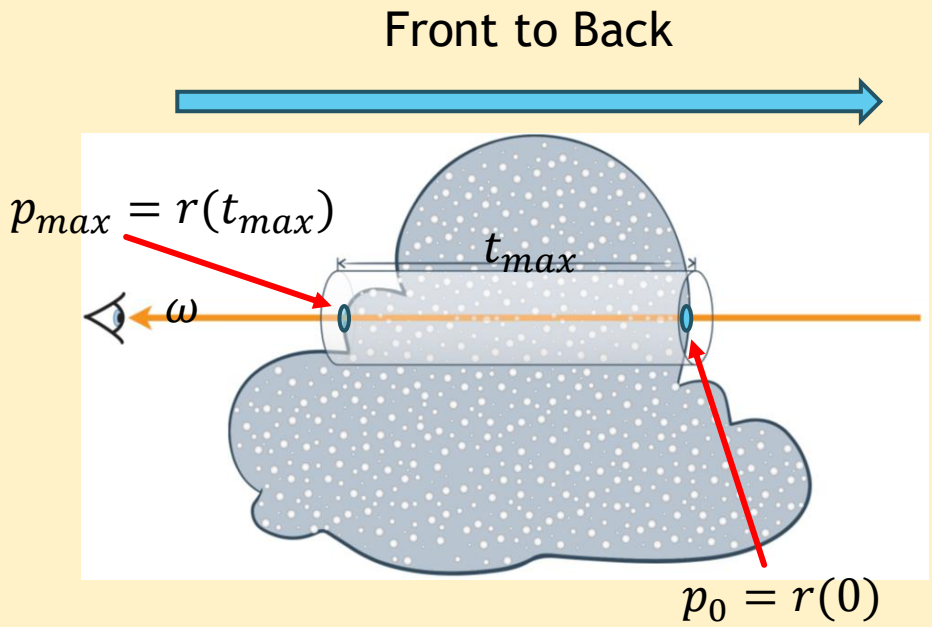
Absorption = 0.15, Ambient light color intensity x50, Heterogeneous volume



Ray Marching with Emission-Absorption Model

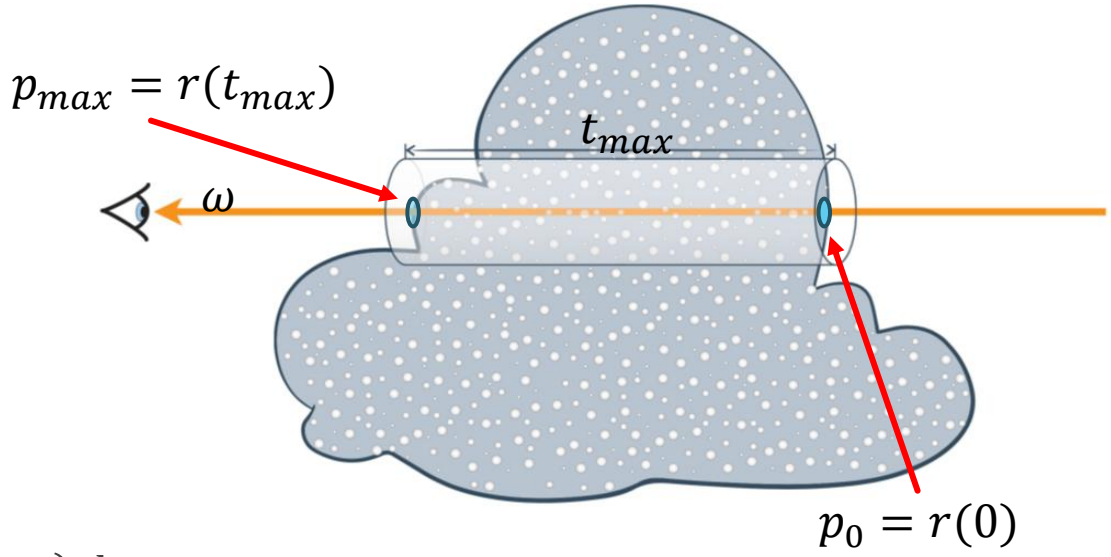
$$L(t) = \int_0^t \mu_a(t')\, L_e(t')\, T(t', t)\, dt' + B\, T(0, t)$$

- Goal: at each position t'
  - Account for L_e and attenuate it at the current position
  - Must also attenuate it all the way till the end of the volume
- Step 1: Write the absorption loop as front to back
- Step 2: Change the loop such as to include emission
  - NOTE: in a front to back approach, at each position along the ray, we can easily retrieve the value of $T(t', t)$



Full Volume Rendering Model

Full Volume Rendering Model Set-Up



- Recall the scattering term

$$L_s(t) = \int_{S^2} f_{\mathbf{x}}(\omega, \omega_i)\, L_i(\mathbf{x}, \omega_i)\, \mathrm{d}\omega_i$$

- And the Radiative Transfer Equation (RTE, full model)

$$\frac{\mathrm{d}L}{\mathrm{d}t}(t) + \mu_t(t)\, L(t) = \mu_s(t) L_s(t) + \mu_a(t) L_e(t)$$

- Also a non-homogeneous 1st order ODE, but with a slightly more complex $q(x)$ due to the inclusion of the scattering effect

$$y'(x) + p(x)\, y(x) = q(x)$$

Volume Rendering Equation

Recall: $L_s(t) = \int_{S^2} f_{\mathbf{x}}(\omega, \omega_i)\, L_i(t, \omega_i)\, \mathrm{d}\omega_i$

Phase function

- The Volume Rendering Equation (VRE) results from solving the EDO of the full RTE
- The resulting expression is given by:

$$L(t) = B\, T(0,t) + \int_0^t \mu_a(t')\, L_e(t')\, T(t',t)\, dt' + \int_0^t \mu_s(t')\, L_s(t')\, T(t',t)\, dt'$$

Source term: does **not** depend on L_i

(In) Scattering term: does depend on L_i (see L_s above)

$$T(t_a, t_b) = e^{-\int_{t_a}^{t_b} \mu_t(t')\, \mathrm{d}t'}$$

Volume Rendering Equation

$$L(t) = B\,T(0,t) + \int_0^t \mu_a(t')\,L_e(t')\,T(t',t)\,\mathrm{d}t' + \int_0^t \mu_s(t')\,L_s(t')\,T(t',t)\,\mathrm{d}t'$$

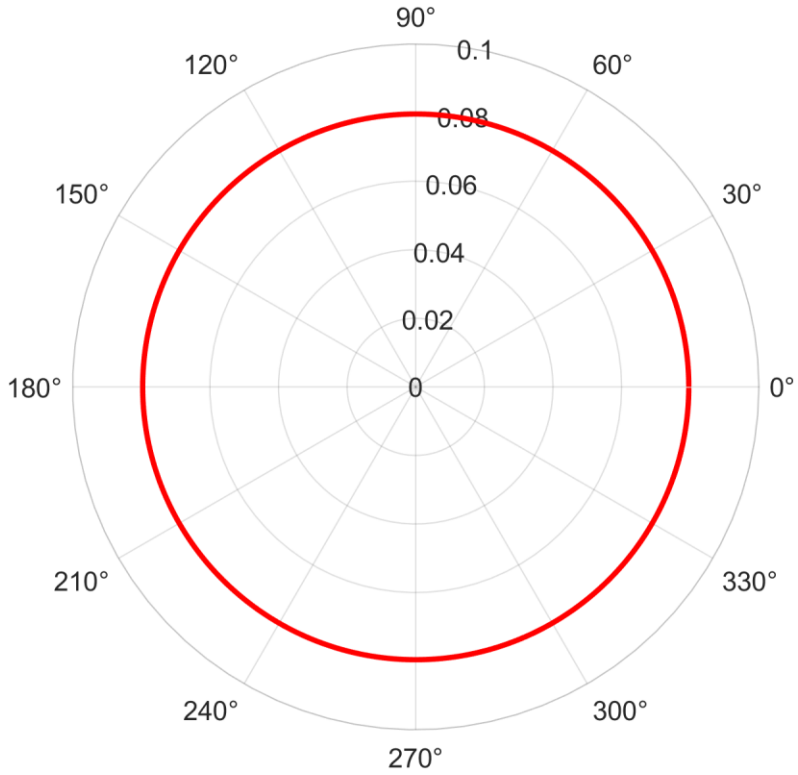
- Note that the integration limits are identic, and that $T(t',t)$ appears on both integrals. We can thus write the above equation using a single integral:

$$\boxed{L(t) = B\,T(0,t) + \int_0^t T(t',t)\left(\mu_a(t')\,L_e(t') + \mu_s(t')\,L_s(t')\right)\mathrm{d}t'}$$

- Note the similarity to the Emission-Absorption VRE previously derived:

$$L^{EA}(t) = B\,T(0,t) + \int_0^t T(t',t)\,\mu_a(t')\,L_e(t')\;dt'$$

Phase Functions



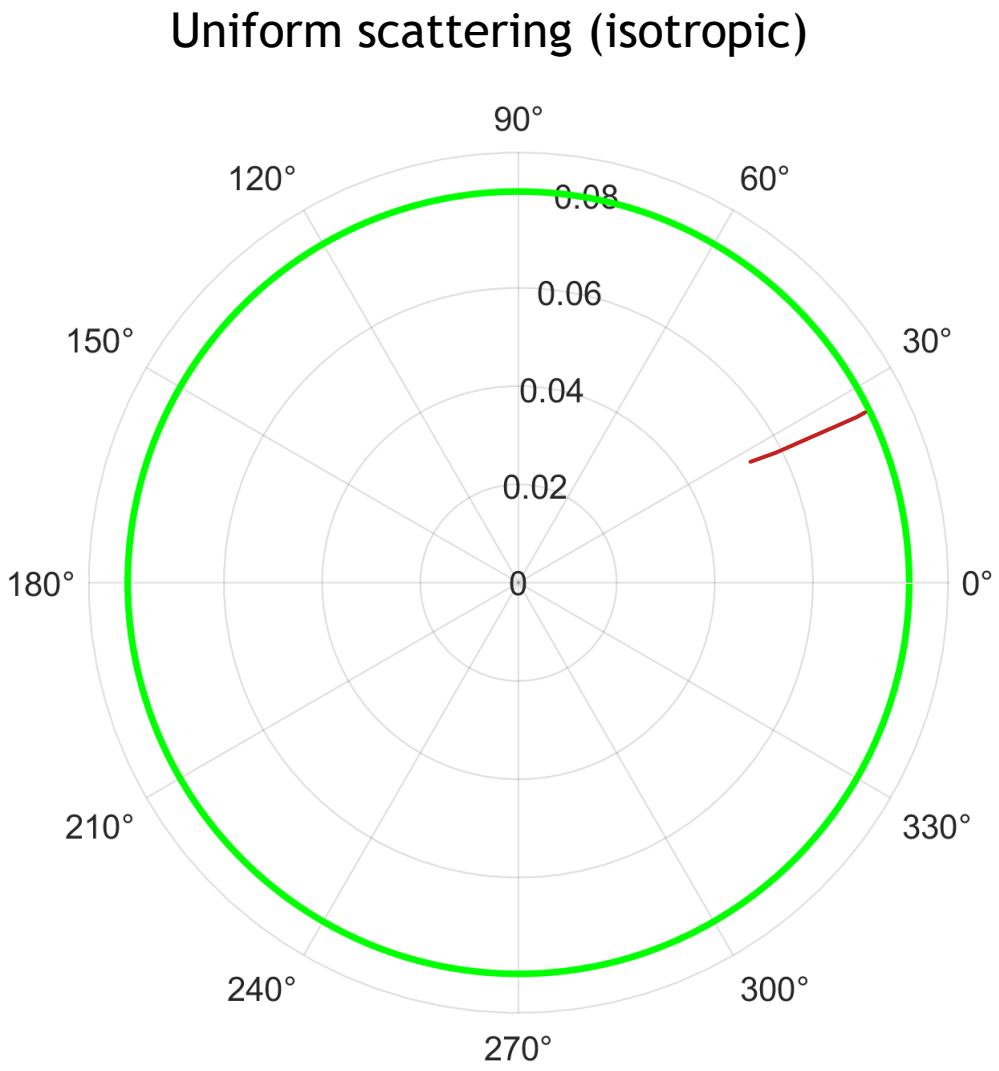
- Simplest
  - Isotropic scattering
  - $f_{\mathbf{x}}(\omega_o, \omega_i) = \frac{1}{4\pi}$
  - Equivalent to diffuse reflection on a surface
- In the polar plot at the right, the angle is measured between ω_o and ω_i



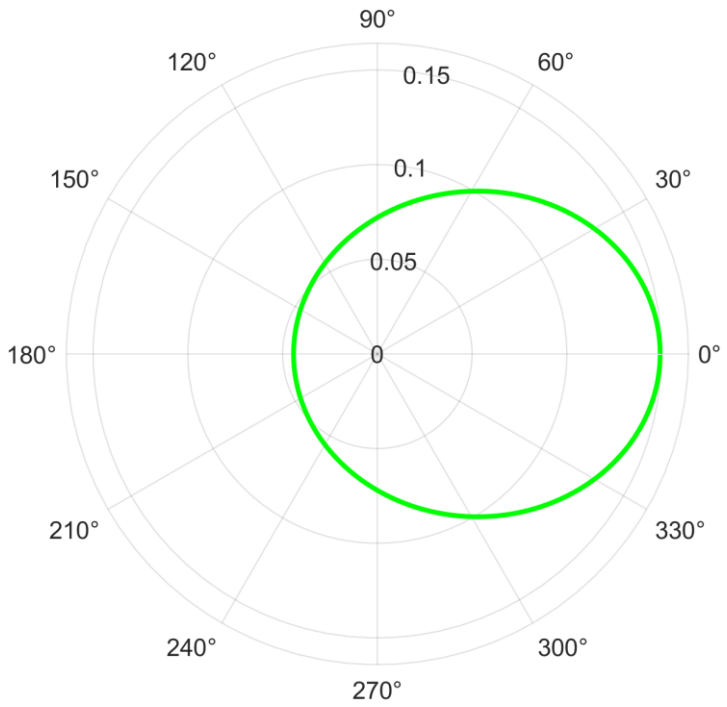
Phase Functions

- Widely used:
  - Henyey and Greenstein (1941)
  - Also empiric, single parameter g (called the asymmetry parameter)
  - $f_{\mathbf{x}}(\omega, \omega_i) = \frac{1}{4\pi} \frac{1-g^2}{\left(1+g^2-2g(\cos\theta)\right)^{3/2}}$

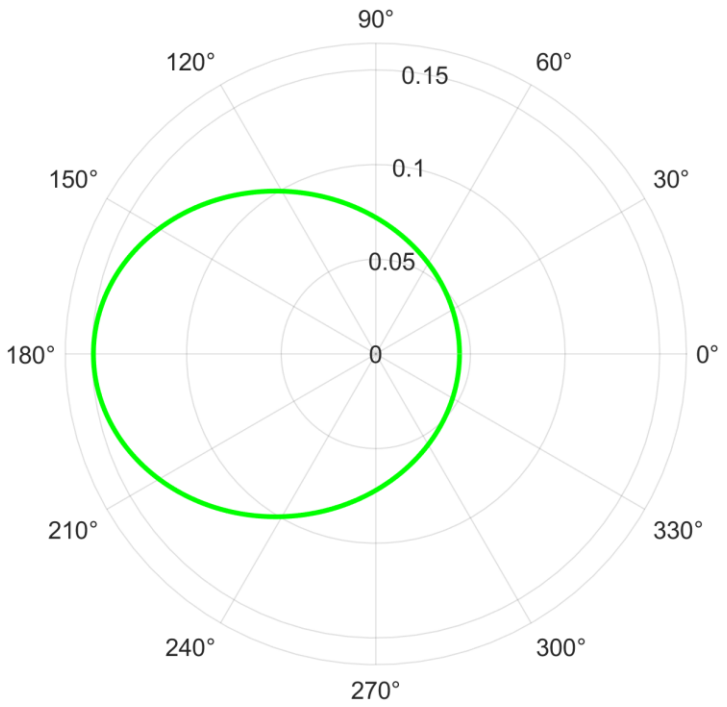
$g = 0$


Uniform scattering (isotropic)
90°
120°
60°
0.08
0.06
150°
30°
0.04
0.02
180°
0
0°
210°
330°
240°
300°
270°

$g = 0.2$ (dominant forward scattering)

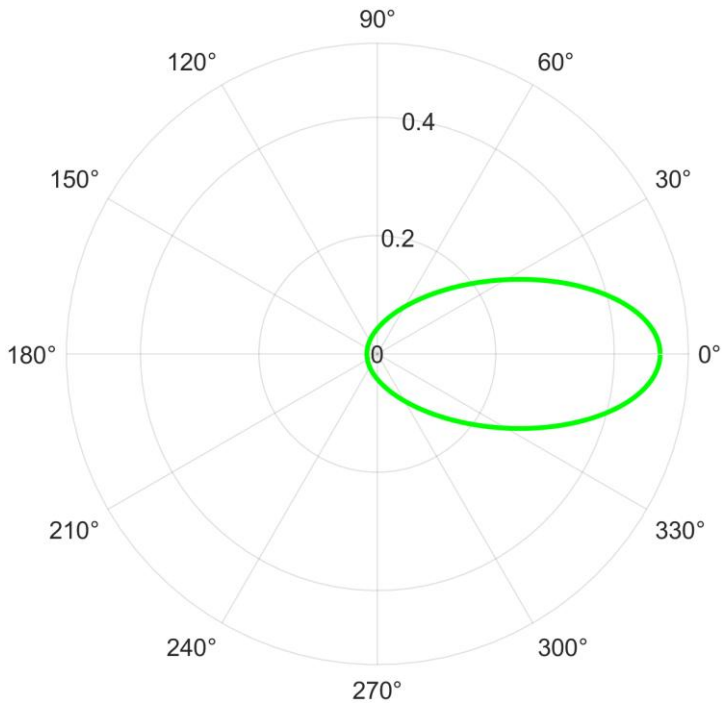


$g = -0.2$ (dominant backwards scattering)

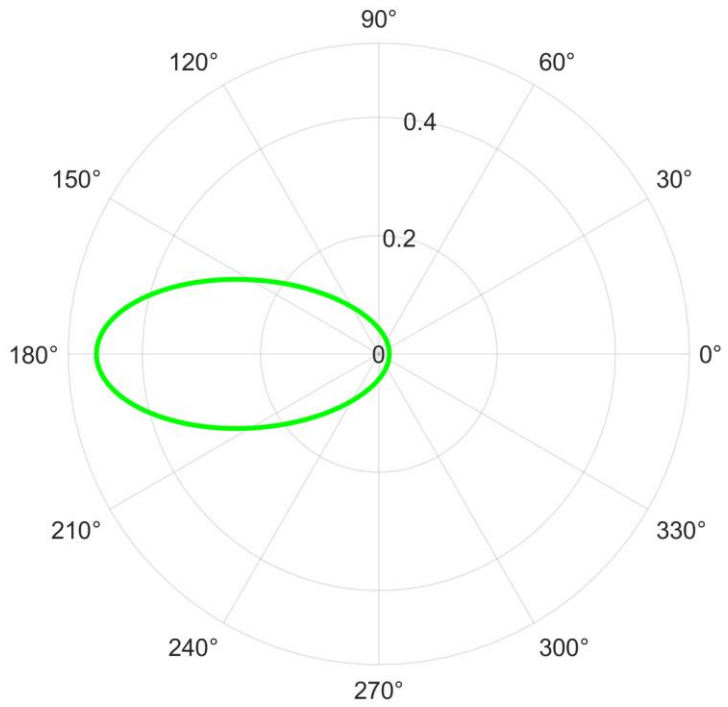


$$g = \pm 0.2$$

$g = 0.5$ (dominant forward scattering)

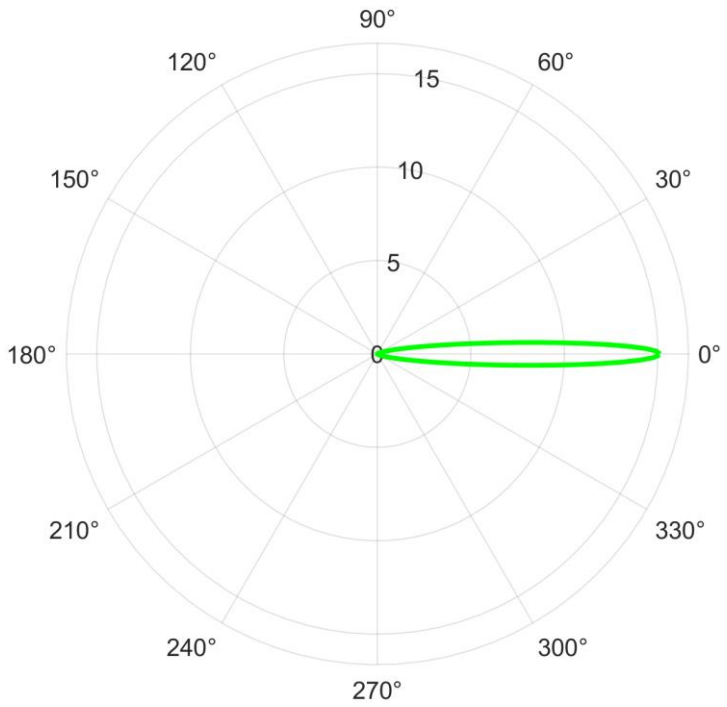


$g = -0.5$ (dominant backwards scattering)

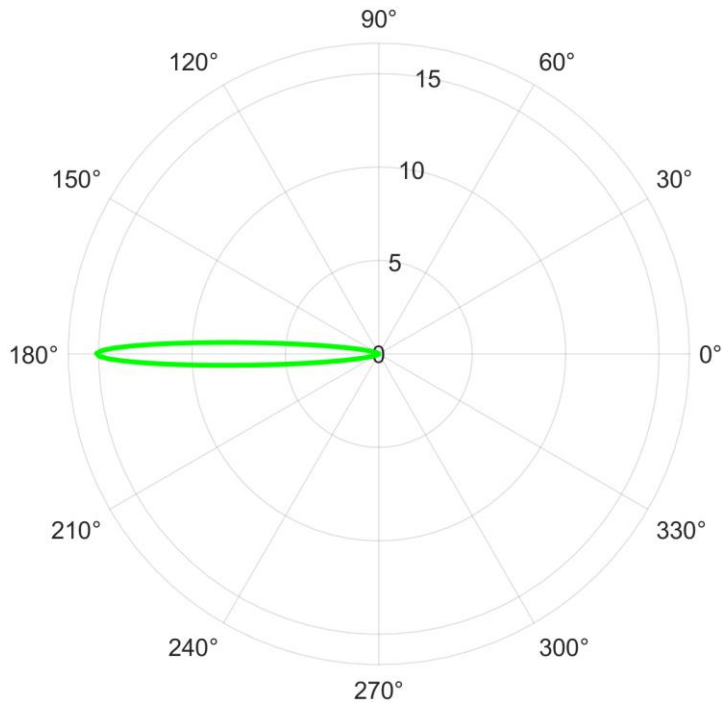


$$g = \pm 0.5$$

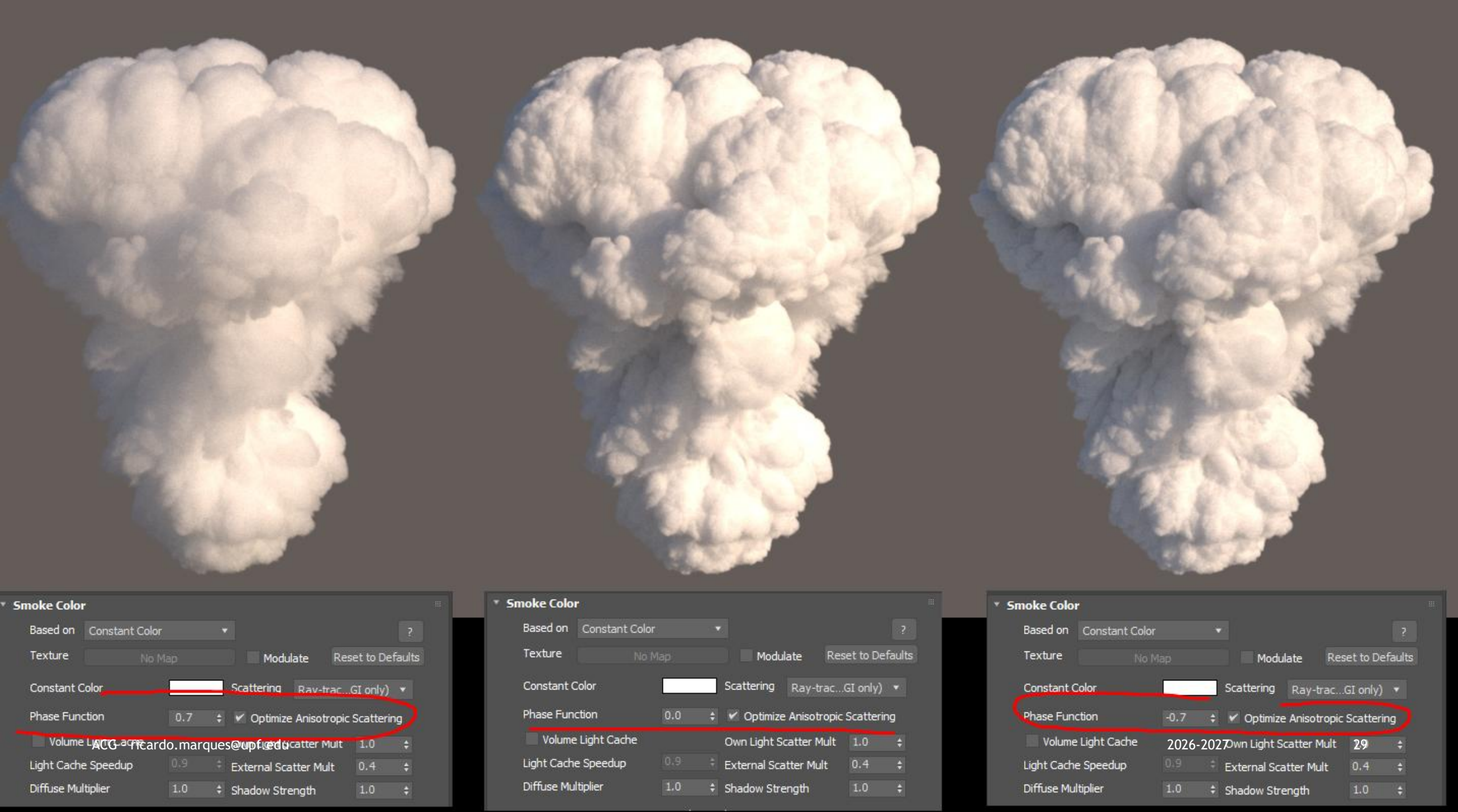
$g = 0.9$ (dominant forward scattering)

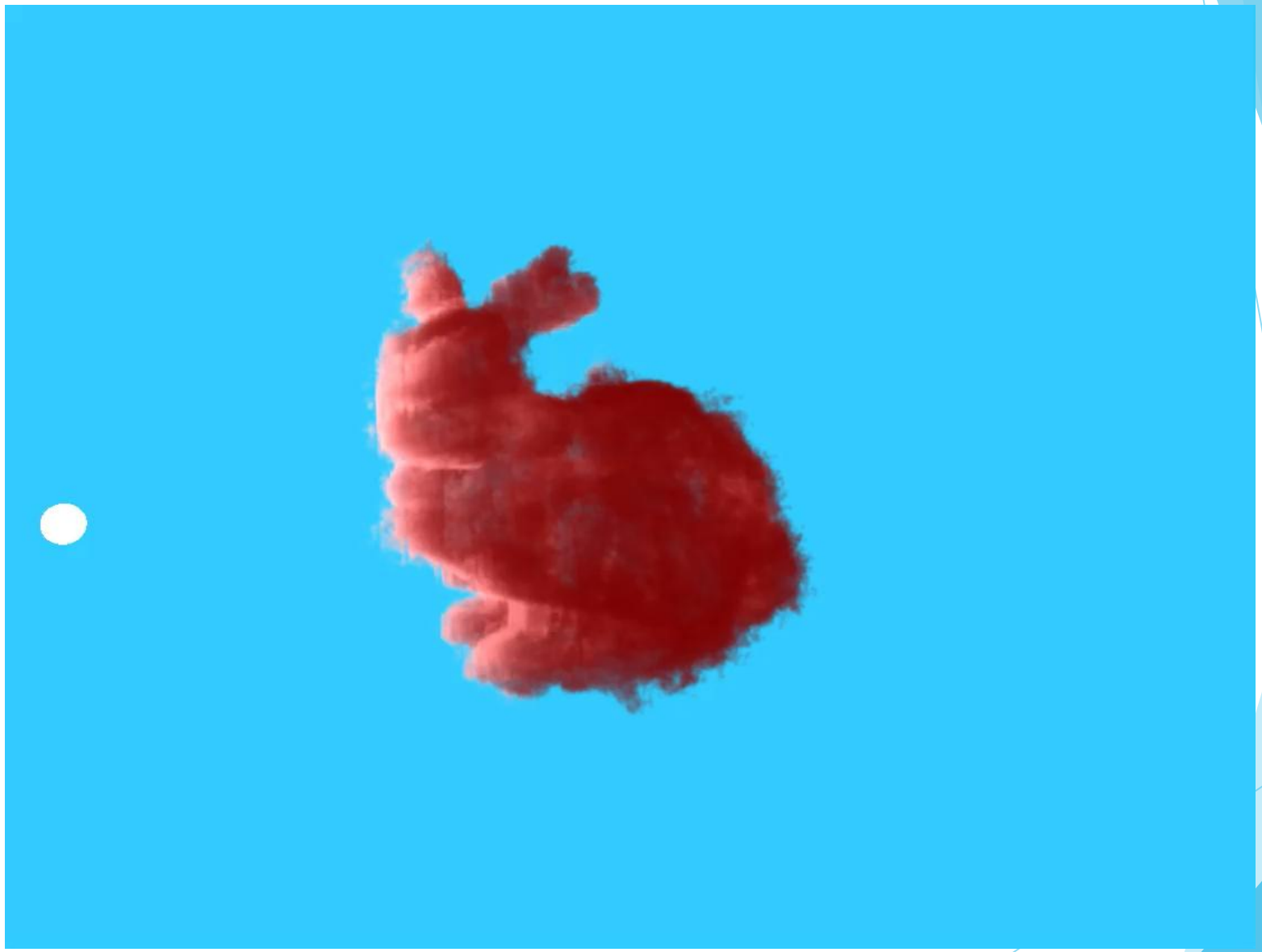


$g = -0.9$ (dominant backwards scattering)



$$g = \pm 0.9$$

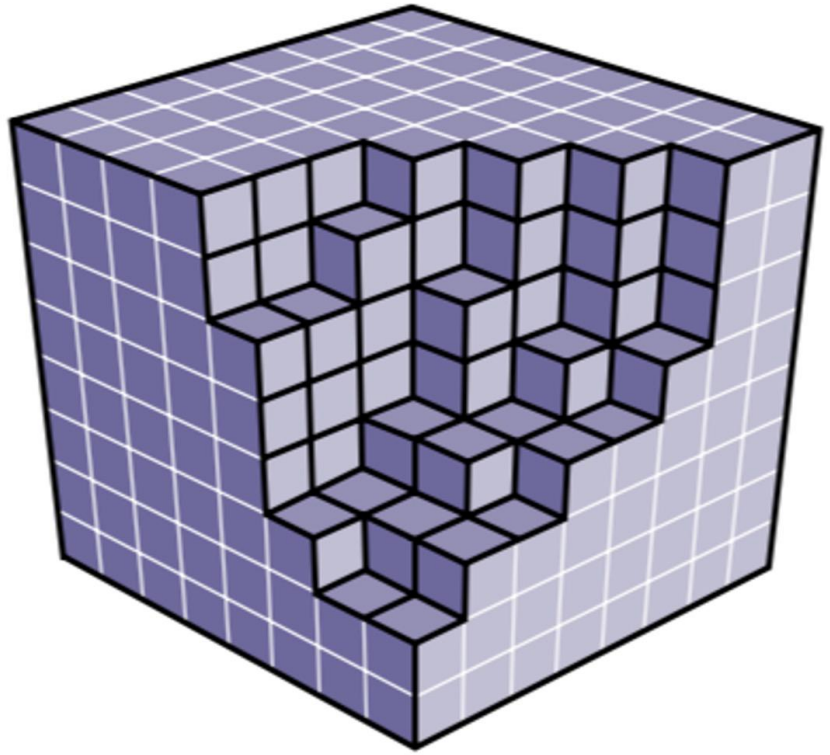

Smoke Color
Based on Constant Color
Texture No Map
Modulate
Reset to Defaults
Constant Color
Scattering Ray-trac...GI only)
Phase Function 0.7
Optimize Anisotropic Scattering
Volume Light Cache
Own Light Scatter Mult 1.0
Light Cache Speedup 0.9
External Scatter Mult 0.4
Diffuse Multiplier 1.0
Shadow Strength 1.0
Smoke Color
Based on Constant Color
Texture No Map
Modulate
Reset to Defaults
Constant Color
Scattering Ray-trac...GI only)
Phase Function 0.0
Optimize Anisotropic Scattering
Volume Light Cache
Own Light Scatter Mult 1.0
Light Cache Speedup 0.9
External Scatter Mult 0.4
Diffuse Multiplier 1.0
Shadow Strength 1.0
Smoke Color
Based on Constant Color
Texture No Map
Modulate
Reset to Defaults
Constant Color
Scattering Ray-trac...GI only)
Phase Function -0.7
Optimize Anisotropic Scattering
Volume Light Cache
Own Light Scatter Mult 1.0
Light Cache Speedup 0.9
External Scatter Mult 0.4
Diffuse Multiplier 1.0
Shadow Strength 1.0



PAUSE

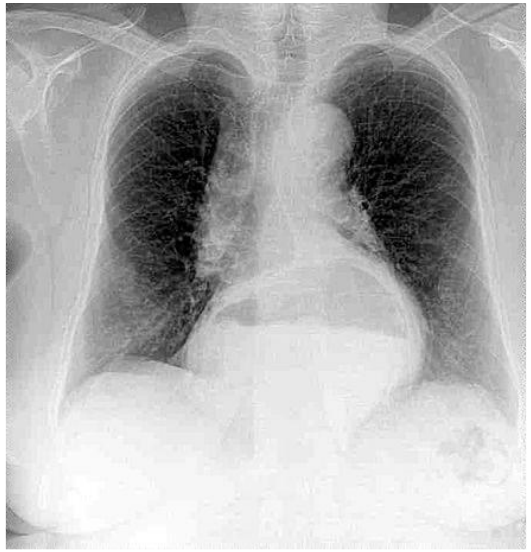
Discretizing Data: Voxels and Grids

Voxels and (Uniform) Grids



- Move from continuous to discrete
- Voxels 3D equivalent of 2D pixels
  - Volume element
- Points which define hexahedral elements (aka cubes)

(Images from Wikipedia)

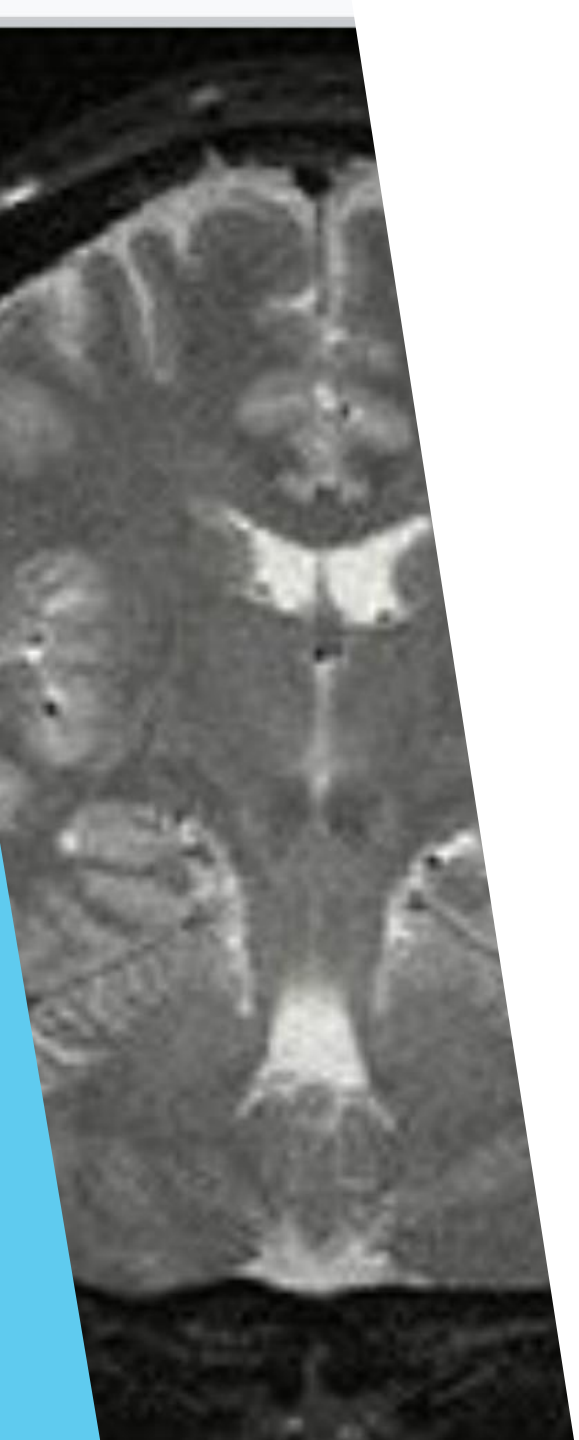




Medical Images: X-Ray and CT Images

- **X-ray images** are 2D images, resulting from absorption of X-rays coming from a source

- **CT images** are slices of Xray images; by using many of them, the result is a '3D' Image (images from Wikipedia)



Medical Images: MRI

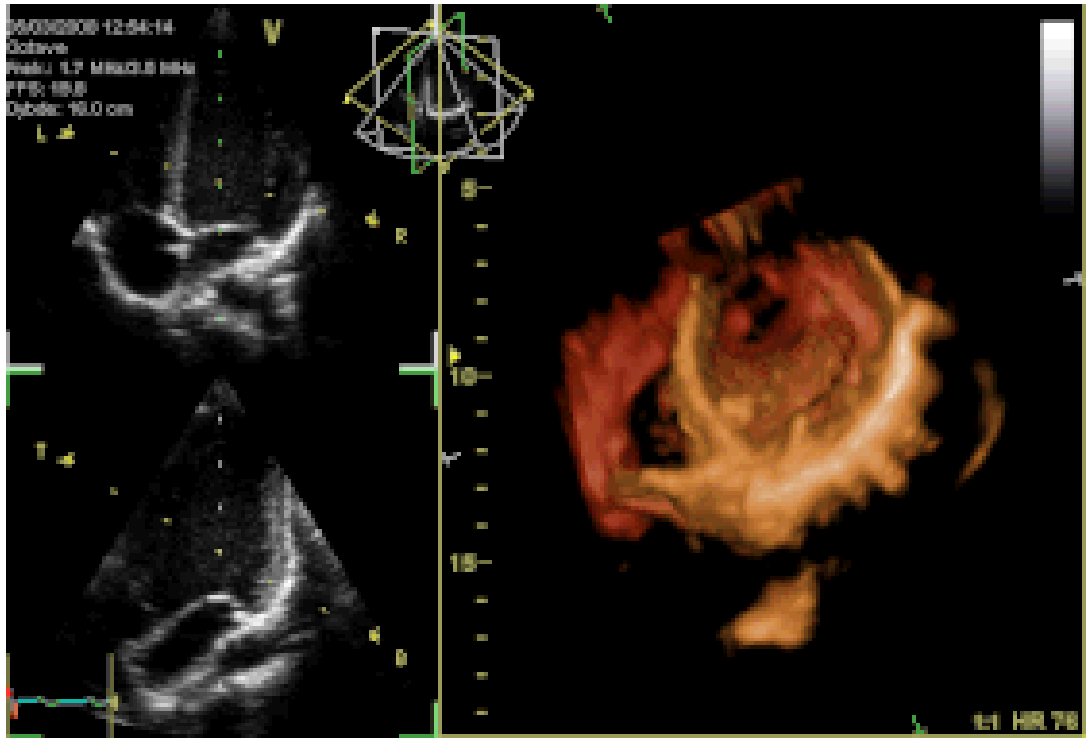
- **MR images** (MRI) obtention is quite different: (very strong) magnetic fields interact with matter, resulting in a 3D response signal which can be measured and stored

- While X-ray and CT images show *structures* (bones, tissues), MR Images show *functionality* (processes). Their combination provides additional information.

- Link Wikipedia

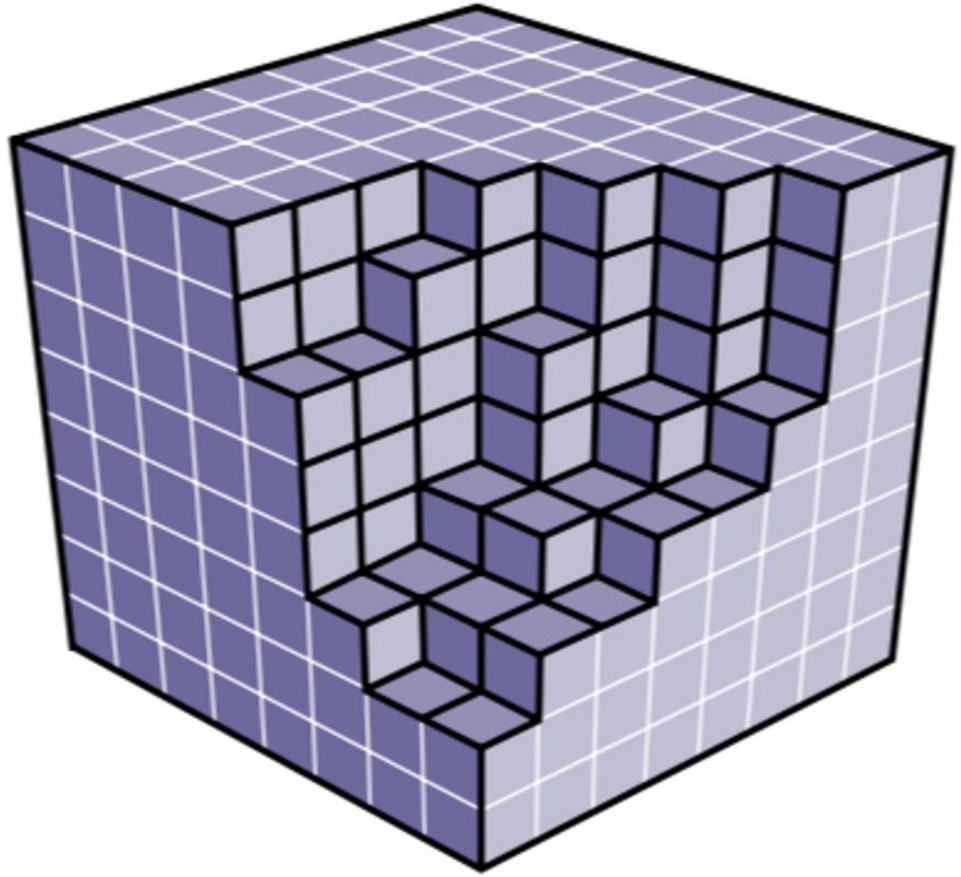
Medical Images: Ultrasound Scans

- The echo of ultrasounds is used to obtain ultrasound scan.
- They can be 2D or ‘3D’



Medical Images: Important Parameters

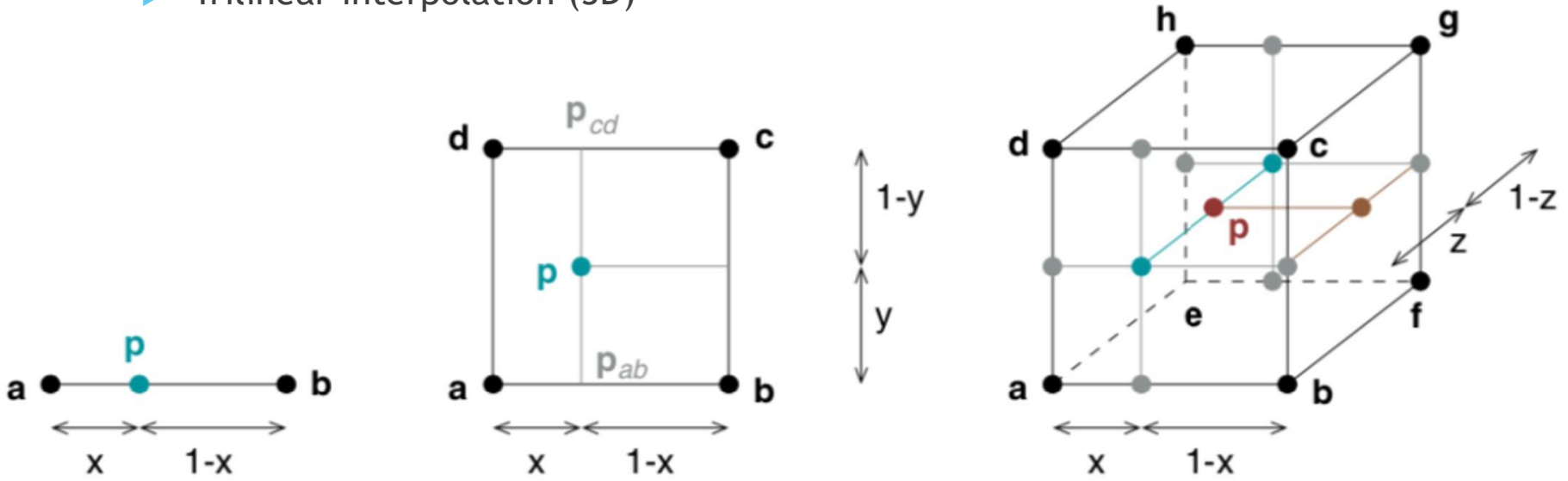
- In images, resolution leads to important parameters.
- For instance, for time varying MRIs:
  - The spatial resolution in each slice
  - The slices resolution related to the distance of two consecutive slices
  - The temporal resolution, how often in time MRIs (or CTs) are obtained.



Interpolation in Uniform Grids

Linear Interpolation

- The most frequent interpolation in CG, when using uniform grids are:
  - Linear interpolation (1D)
  - Bilinear interpolation (2D)
  - Trilinear interpolation (3D)

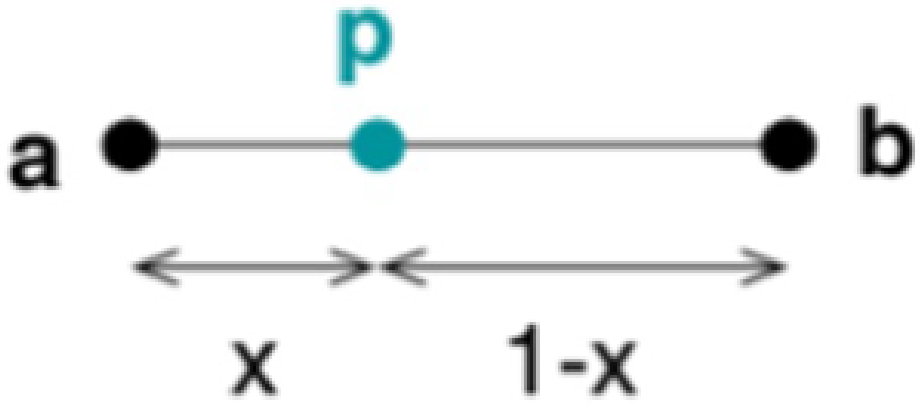


Linear interpolation

- Given two locations a and b such that $|b - a| = 1$
- Given the function values $f(a)$ and $f(b)$
- Goal: estimate the function value at p

$$f(p) \approx (1 - x)f(a) + x\, f(b)$$

- x should be understood as a weighting factor which weights the values of a and b in the sum



Linear Interpolation in Regular Grids

- Let i be the index of the vertexes in a 1D regular grid, and s be the distance between grid points
- Then, for a point p such that $v_i \leq p \leq v_{i+1}$, the linear interpolation can be expressed as:

$$f(p) \approx (1 - x)\, f(v_i) + x\, f(v_{i+1})$$

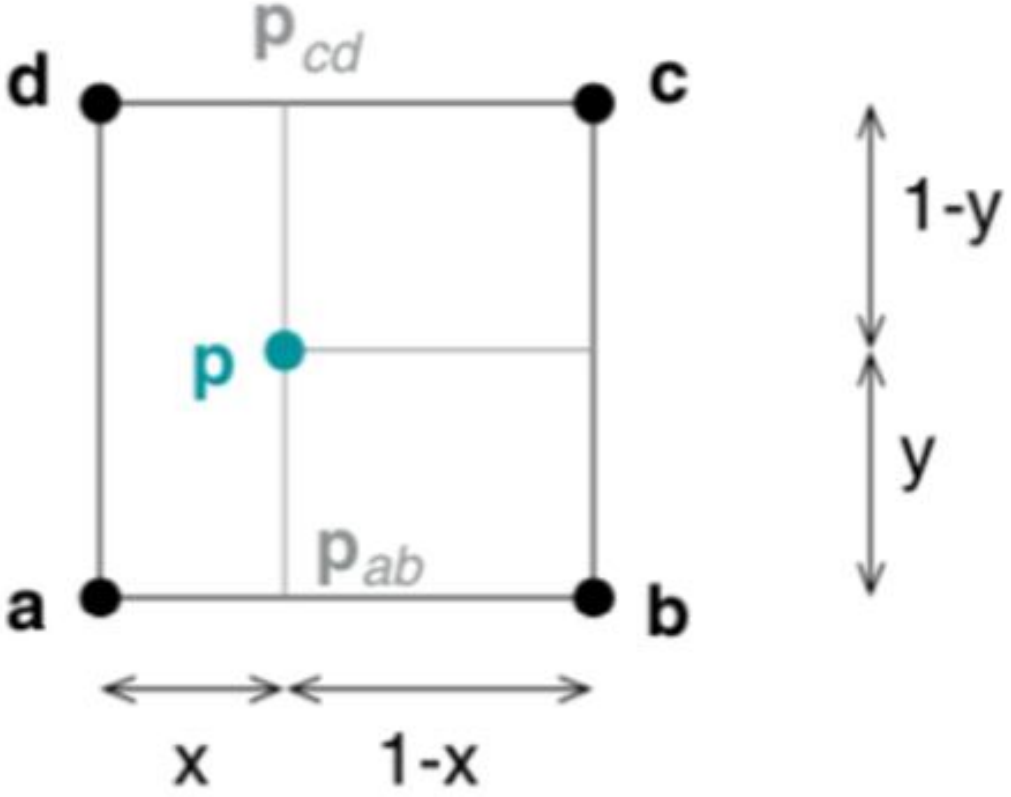
where $x = \frac{d}{s}$, with d being the distance between v_i and p

v_i (closest smaller vertex) $\rightarrow d = p - v_i \geq 0$

- *Question: How would you compute i?*

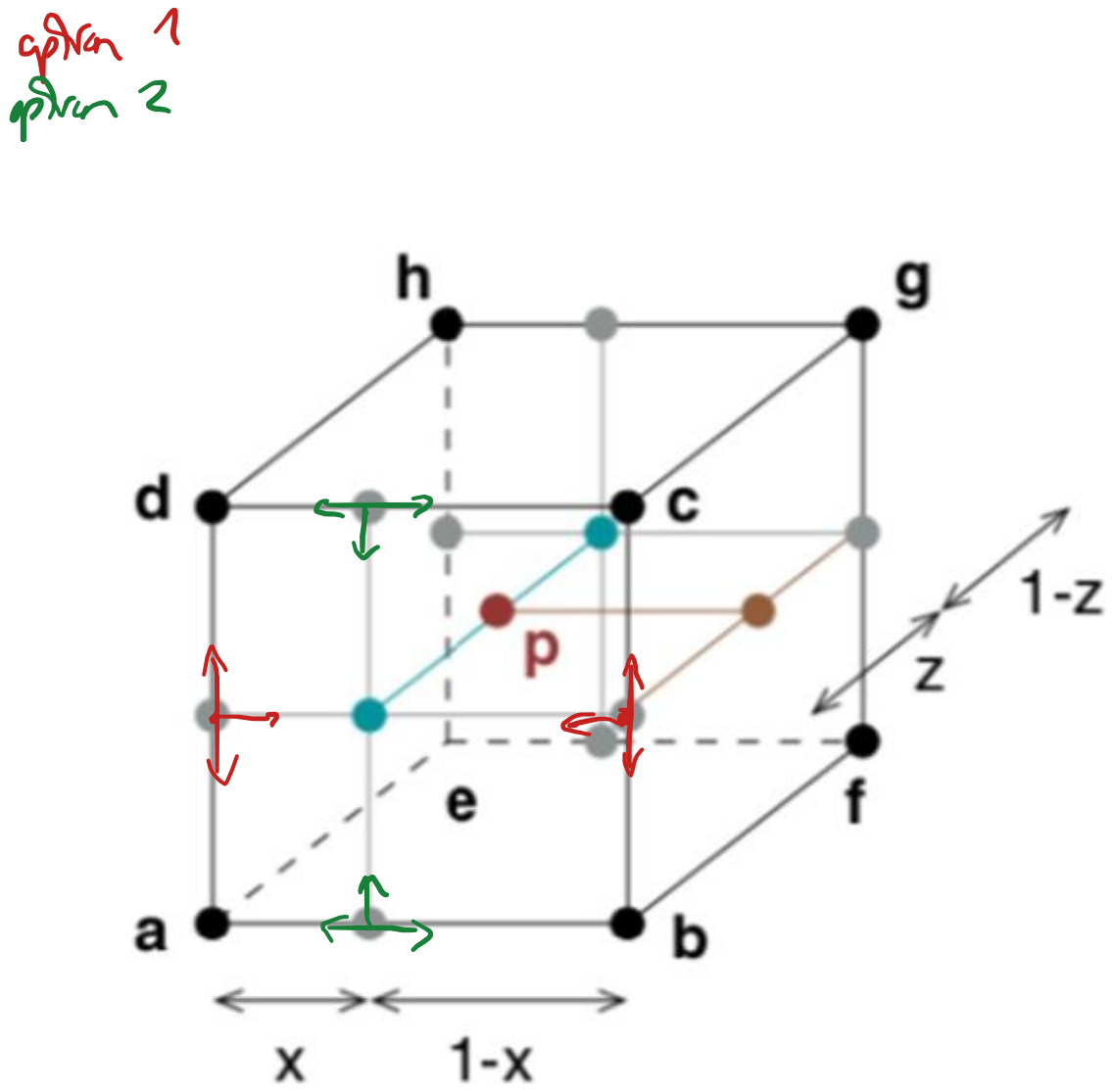
Bilinear Interpolation

- Used for bi-dimensional data
- Consists of:
  1. Obtaining $f(p_{ab})$ and $f(p_{dc})$ via *horizontal* linear interpolation,
  2. Perform a linear *vertical* linear interpolation of $f(p_{ab})$ and $f(p_{dc})$



Trilinear Interpolation

- Similarly, the trilinear interpolation consists of:
  1. Performing a bilinear interpolation on the square [a,b,c,d]
  2. Performing a bilinear interpolation on the square [e,f,g,h]
  3. Linear interpolate the results of the two linear interpolations above
- Directly available in the GPU



Exercise Trilinear Interpolation

- Consider a 3D uniform grid, where consecutive points are 8 units apart, so that the length of the side of each cubic voxel is 8 units. Consider two points $p = (14, 17, 23)$ and $q = (18, 26, 5)$.
  1. Determine the vertexes of the voxels to which p belongs.
  2. Determine the vertexes of the voxels to which q belongs.

- Let $v_{i,j,k}$ be the vertex with smaller indexes among the vertexes of the voxel containing p. Let also:
  - $\mu_a(v_{i,j,k}) = 0.01$; $\mu_a(v_{i+1,j,k}) = 0.02$; $\mu_a(v_{i,j+1,k}) = 0.015$ and $\mu_a(v_{i+1,j+1,k}) = 0.03$
  - $\mu_a(v_{i,j,k+1}) = 0.05$; $\mu_a(v_{i+1,j,k+1}) = 0.025$; $\mu_a(v_{i,j+1,k+1}) = 0.04$ and $\mu_a(v_{i+1,j+1,k+1}) = 0.035$

  3. Using a trilinear interpolation, approximate the value of $\mu_a(p)$

Credits

- Thank you to Josep Blat for making available the slides for ACG (2023)
- Thank you Víctor Ubieto for the images with Volumes with our framework
- Several slides and images are from the PBRT Book

Calendar

Week no.	Date	Theory	Practice	Seminar	Deliveries
Week 1	28 Sep 02 Oct	**T1**: Introduction	**P1**: Introduction to A1.1	-	
Week 2	05 Oct 09 Oct	**T2**: Basic Ray Tracing		**S1**: Work on A1	
Week 3	12 Oct 16 Oct	**T3**: Ray Casting, Recursive Ray Tracing	**P2**: Introduction to A1.2		
Week 4	19 Oct 23 Oct	**T4**: Rendering Equation, Monte Carlo Integration and Direct Illumination **T5**: Indirect Illumination and Path Tracing		**S2**: Work on A1	**D.A1**
Week 5	26 Oct 30 Oct	**T6**: Volume Rendering: Absorption-Only Model	**P3**: Introduction to A2.1		
Week 6	02 Nov 06 Nov	**T7**: Volume Rendering: Emission, Absorption and Scattering Model **T8:** Neural Radiance Fields and Topics' Presentation		**S3**: Work on A2	
Week 7	09 Nov 13 Nov	**T9**: **Midterm**	**P4**: Introduction to A2.2		
Week 8	16 Nov 20 Nov		**P5**: Work on A2	**S4**: Work on A2	**D.A2**
Week 9	23 Nov 27 Nov	**T10**: **1st Presentation**	**P6**: Work on FP	-	
Week 10	30 Nov 04 Dec	**T11**: **2nd Presentation** **(First Session)** **T12**: **2nd Presentation** **(Second Session)**		**S5**: Work on FP	**D.FP**



The End

Questions?